

Features, Labels, Targets

Project: Responsible Plant Classifier

1. Features, Labels, Targets

Teach the learning cycle

Primary target: Separate prediction inputs from observed targets

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

The ML Lifecycle

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish fitting a rule from applying a fixed rule

Reference example, not the only acceptable wording: The input is width x and the label is the observed category 0 or 1. Candidates are 2, 4 and 6. Selection uses mistakes on the four training cards only; held out labels must not guide it.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate input set.

Concept to explain

Features are the information available at prediction time; the target is the value the model should estimate; labels are observed target values in examples.

Term · Meaning

Term: feature

Meaning: input variable used for prediction

Term: target

Meaning: quantity or category to estimate

Term: label

Meaning: observed target value in a dataset

Term: row

Meaning: one example with fields

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach features, labels and prediction time in two 45-minute sessions using fictional records. Students must define the decision moment before judging columns. The key distinction is that labels may be known for historical training but must not be fed into predictions for new cases. This lesson prepares inputs; it does not demonstrate a trained model or establish that loan prediction is useful. Use paper records or local Python, without accounts or personal borrowing data.

Readiness answer

They are reserved to evaluate the fixed choice. Using their labels to choose it makes them part of development.

Printed lesson and workbook

Two cards labelled checkout and later return

Optional Python 3 editor; no external packages

Blank field meaning unit availability role table

End goal for the pictorial case

By the end, I can explain features, labels, targets using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Mark features and targets in three datasets and reject information unavailable at real prediction time.

MODEL IDEA Features are the information available at prediction time; the target is the value the model should estimate; labels are observed target values in examples.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Plant row: leaf_width=6 cm, color_score=0.72, health=spotted. Features are leaf_width and color_score because they exist before prediction.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Plant row: leaf_width=6 cm, color_score=0.72, health=spotted.

STEP 2 Features are leaf_width and color_score because they exist before prediction.

STEP 3 Target is health; the stored value spotted is this row's label.

STEP 4 A treatment-result field recorded later is not a valid prediction-time feature.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

Rows contain leaf length, edge type and healthy/spotted label.

Identify features, target and prediction-time inputs.

Exact explanation to model

Length and edge type are candidate features; the label is the target. Do not supply the true target as an input to predicting it. Evaluate feature usefulness.

Explain the added worked example

Week 1 separated training examples from testing examples. This week asks a different question: could each input actually be known when a prediction is needed? A historical table may contain facts added after the event. A model trained with those facts can appear successful even though the inputs will be missing when it is used.

Use fictional library loans. At the instant a book is checked out, predict whether it will be returned more than 14 days later. For completed historical loans, label 1 means days_to_return is greater than 14; label 0 means it is at most 14. Exactly 14 days is on time. Assume no changed due dates in this exercise. All records and values are invented; do not collect classmates' borrowing histories.

queue_at_checkout is a saved count of waiting requests at that instant. It is not the queue when we later export the table. Candidate means available to consider, not proven useful or appropriate. Exclude the arbitrary loan identifier from this exercise; keep it separately to track records.

The feature order is always pages first, queue count second. The first output is [120, 2]; the separate label is 1 because this historical loan took 18 days. The new loan produces [80, 0] even though its return outcome does not yet exist. The same feature function serves past and new records. It deliberately selects allowed fields rather than copying every column except returned_late.

This code prepares inputs; it does not fit a model or predict lateness. It assumes the two feature fields

contain valid nonnegative integer values and pages is positive. A missing required field raises `KeyError`. A real pipeline needs explicit validation and a missing-data policy.

Key terms

Term · Meaning

Term: Feature contract

Meaning: A specification of input meaning units order and timing

Term: Target

Meaning: The outcome a prediction aims to estimate

Term: Prediction time

Meaning: The moment when the prediction must be made

Term: Target leakage

Meaning: Predictor information that reveals the outcome but is unavailable at prediction time

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

For every proposed input, ask what it means, when it was recorded, and whether the same definition can be supplied at checkout. A field in the historical export is not automatically available at prediction time. An input calculated from later information is also unavailable, even if its name does not mention the outcome.

A completed loan has pages 200, queue_at_checkout 3 and days_to_return 14. Write the feature list and label. Someone adds overdue_fee calculated after return. Can it become a feature for the checkout prediction? Explain.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Features, Labels, Targets

Choose features that exist when a plant prediction must be made.

The facts we will use

Predict next-week wilting at today's photo capture. Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the target recorded later.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Features, Labels, Targets

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Mark prediction time

The decision occurs today, before next week's outcome.



2. Select inputs

Today's light and soil readings are available features; plant_id identifies the plant.



3. Keep target separate

next_week_wilt supplies the training label, not an input available today.

CHANGE ONE THING

A table adds tomorrow_soil_moisture as an input.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Mark prediction time

What happens and why: The decision occurs today, before next week's outcome.

Step: 2. Select inputs

What happens and why: Today's light and soil readings are available features; plant_id identifies the plant.

Step: 3. Keep target separate

What happens and why: next_week_wilt supplies the training label, not an input available today.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. next_week_wilt supplies the training label, not an input available today. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Eligible fields are not automatically useful or causal. Measure their predictive value with a valid split.

Change one thing in the pictured case

Change: A table adds tomorrow_soil_moisture as an input.

Prediction: Reject it for this prediction-time contract.

Reason: Tomorrow's measurement is unavailable when today's prediction must be issued.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Rows contain leaf length, edge type and healthy/spotted label.

Write the question to explain: Identify features, target and prediction-time inputs.

Reveal after predictions: Length and edge type are candidate features; the label is the target. Do not supply the true target as an input to predicting it. Evaluate feature usefulness.

Keep for an independent attempt: A row contains leaf_length_cm=8, leaf_width_cm=3 and species=fern. For predicting species, identify two features and the target.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: the ordered inputs are [200, 3] and the observed label is 0 at exactly 14 days. The fee is calculated later and cannot be used for an earlier checkout prediction. Exit example: use [pages, queue_at_checkout], in pages and requests, both recorded at checkout. An unknown future return outcome is not evidence of an on-time return and must not be silently labelled 0.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Recall why held out labels must not choose the Week 1 threshold. Connect separate examples to the separate question of input availability.

Stage: 7 to 18 minutes

Teacher action: State the checkout moment and label boundary. Place each field on the timeline. Explain why an arbitrary identifier can stay as metadata without becoming an input.

Stage: 18 to 32 minutes

Teacher action: Trace the feature function on past and new records. Complete W1 and W2. Require the correct order and the 14 versus 15 day distinction.

Stage: 32 to 45 minutes

Teacher action: Complete the guided 200-page loan and fee question. Ask which fields are unknown on a new checkout and why unknown does not mean zero.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the contract: names, units, order and availability time. Ask students to explain one later field without using only the word leakage.

Stage: 8 to 20 minutes

Teacher action: Use W3 to challenge a perfect test score. Trace how days_to_return reveals the answer even on separate test rows.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently. Require inputs, target, exclusions, boundary answer and unknown-outcome explanation before giving hints.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit contract. Use the fresh retry after support where needed and record the support given.

Reduce help deliberately

Completion task: A completed loan has pages 96, queue_at_checkout 4 and days_to_return 14. Give the ordered feature list and target. Then state what changes if it took 15 days instead.

Teacher check: The feature list is [96, 4]. At 14 days the target is 0. At 15 days the target becomes 1; the checkout feature list remains [96, 4].

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Features labels and prediction time. Proposed explanation: “Every stored column is a feature”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A historical export combines information from different times. Require a meaning and timestamp before treating a column as an input.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why can a historical table contain a field that is an invalid input? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

Historical exports combine measurements from different times. Availability at prediction time, not table presence, determines eligibility. Changed-case check: Reject it for this prediction-time contract. Tomorrow's measurement is unavailable when today's prediction must be issued.

Changed-case reasoning: Reject it for this prediction-time contract. Tomorrow's measurement is unavailable when today's prediction must be issued.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Feature-or-Target Sort: annotate three tables, explain availability time, and find one disguised post-outcome field.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Any column correlated with the answer is automatically a good feature.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Every stored column is a feature: A historical export combines information from different times. Require a meaning and timestamp before treating a column as an input.

Removing the label column removes leakage: days_to_return and an outcome-based fee may still disclose the target. Audit how every field was created.

An unknown outcome is a negative label: Unknown means not established. In this exercise labels come from completed historical returns. Do not turn unavailable outcomes into on-time labels.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed

incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Length and edge type are candidate features; the label is the target. Do not supply the true target as an input to predicting it. Evaluate feature usefulness.

Independent question

A row contains leaf_length_cm=8, leaf_width_cm=3 and species=fern. For predicting species, identify two features and the target.

Expected independent answer

Leaf length and leaf width are features; species is the target label. Species must not also be used as an input for this prediction task.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “The target can be included as an input because it improves apparent accuracy.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Verified species is the target and must not be supplied as a prediction feature. Use length and edge type if available at prediction time, then evaluate their usefulness. Numeric encoding of the target does not make it a legitimate input.

Explanations for the additional workbook examples

W1 Assign the roles

For the worked loan table, identify the two candidate features, the target label and the tracking field. Explain why `days_to_return` is not a predictor.

Candidate features are `pages` and `queue_at_checkout`. The target is `returned_late` and the tracking field is `loan_id`. `days_to_return` is known only after return and directly defines the target, so it must not enter checkout predictions.

W2 Trace the feature contract

A completed loan has `pages` 96, `queue_at_checkout` 4 and `days_to_return` 14. Give the ordered feature list and target. Then state what changes if it took 15 days instead.

The feature list is `[96, 4]`. At 14 days the target is 0. At 15 days the target becomes 1; the checkout feature list remains `[96, 4]`.

W3 Repair a misleading score

A model has every test prediction correct after using `pages` and `days_to_return`. A learner says a separate test set makes this valid. Explain the flaw and repair the inputs.

The test rows contain future information that reveals the label. Separate rows do not make `days_to_return` available at checkout. Remove later outcome fields, use the checkout feature contract, and rerun development and evaluation with honest records.

W4 Audit a new task independently

A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes. Fields are `route_steps` and `queue_at_dispatch` captured at dispatch, `trip_id` for tracking, `arrival_minutes` measured after arrival, and `slow = 1` when `arrival_minutes > 10`. Specify inputs, order, target and exclusions. For a past trip with steps 30, queue 2, arrival 10, give features and label. For a new trip with steps 24 and queue 0, give features and explain whether `slow` is known.

Use `[route_steps, queue_at_dispatch]` in that order, measured in steps and queued trips at dispatch. `slow` is the target. Keep `trip_id` as tracking metadata; exclude `arrival_minutes` and `slow` from inputs. Past features are `[30, 2]` with label 0 because 10 is not greater than 10. New features are `[24, 0]`; `slow` is unknown until the outcome can be established.

W5 Handle an unavailable input

For the robot task, a dataset stores `queue_now` at export time but has no saved `queue_at_dispatch`. Can you rename that field to make it usable? State a defensible next step.

Renaming does not restore the missing dispatch-time value. Recover a reliable timestamped dispatch snapshot if available, or remove the feature and reevaluate the revised design. Do not invent a zero queue or copy the later count.

Assess independent transfer

Selected case: A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes. Fields are route_steps and queue_at_dispatch captured at dispatch, trip_id for tracking, arrival_minutes measured after arrival, and slow = 1 when arrival_minutes > 10. Specify inputs, order, target and exclusions. For a past trip with steps 30, queue 2, arrival 10, give features and label. For a new trip with steps 24 and queue 0, give features and explain whether slow is known.

Selected case answer

Use [route_steps, queue_at_dispatch] in that order, measured in steps and queued trips at dispatch. slow is the target. Keep trip_id as tracking metadata; exclude arrival_minutes and slow from inputs. Past features are [30, 2] with label 0 because 10 is not greater than 10. New features are [24, 0]; slow is unknown until the outcome can be established.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Define the Plant Classifier target and a first feature list; record units, allowed values, collection method, and availability time.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 3: Classification or Regression?. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Place field cards before or after the prediction-time line. Supply an empty contract table, not completed answers. Extension: a queue measurement is recorded before checkout but overwritten at export. Explain why its current value is not necessarily the original snapshot, and propose retaining timestamped history.

Require W4 features [30, 2], target 0, new features [24, 0], unknown new target, and the timing reason for exclusions. A column list without timing justification is insufficient. Fresh retry: at print-job submission, predict whether completion takes more than 5 minutes. Inputs are page_count then queue_at_submit. A completed job has 12 pages, queue 1, completion_minutes 6: inputs [12, 1], label 1. A new job has 8 pages, queue 0: inputs [8, 0], label unknown. Exclude completion_minutes and the label from inputs; keep job_id as metadata. Exactly 5 minutes would have label 0.

Week 3 distinguishes classification targets from regression targets. Carry forward the explicit target definition and prediction time. Teacher reference: https://scikit-learn.org/stable/common_pitfalls.html#data-leakage (reviewed 2026-09-26).

Plan the delayed return

Suggested return points in this level: ai-4/3, ai-4/5, ai-4/8. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier

independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.